

Performance Benchmarking of Embedded Battery State Estimation: A Measurement-Only Comparison of Direct-Measurement, Hybrid, and Data-Driven Estimators

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Abstract

This paper presents a measurement-only robustness benchmark for battery State-of-Charge (SOC) estimation under realistic Battery Management System constraints. Four estimator classes are compared: direct measurement by Coulomb counting (DM), SOH-adaptive Coulomb counting (HDM), a second-order equivalent-circuit model with extended Kalman filtering (HECM), and a GRU-based data-driven estimator (DD). Model development and evaluation are cell-disjoint. The final benchmark covers 16 predeclared 24-h windows from six independent Lithium Iron Phosphate holdout cells, 19 nominal and disturbed-measurement cases, and 6,720 causal model runs. Randomized scenarios use fixed repeated seeds; cells, rather than time samples, are treated as independent units in a hierarchical bootstrap with 10,000 repetitions. Under nominal measurements, DD provides the lowest cell-macro error (MAE 0.0258, RMSE 0.0300), followed by HECM, HDM, and DM. The adverse signed current-gain penalty grows monotonically for all estimators; at 3%, the MAE increase is 0.0107 for DM, 0.0100 for HDM, 0.0059 for HECM, and 0.0038 for DD. HECM corrects an imposed 10-percentage-point initialization mismatch fastest on aver-

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age, whereas DD combines comparatively strong initialization and post-dropout recovery. Missing samples and burst dropout are more consequential than timing jitter, and voltage spikes reveal a local DD transient that is largely hidden by full-window MAE. A paired reference-SOH ablation shows that replacing the causal LSTM-SOH trace does not uniformly improve SOC error, emphasizing that the estimators were calibrated for their deployed feature pipeline. Finally, isolated SOC-core tests on an STM32H753ZI verify close software–hardware agreement and quantify latency and static memory. The results support scenario-specific model selection while limiting conclusions to the tested LFP cells, operating envelope, models, and implementation protocol.

Keywords: Battery management system, State of charge, State of health, Performance benchmark, Direct measurement, Equivalent circuit model, Data-driven estimation, Embedded artificial intelligence

1 Nomenclature

2 *Acronyms*

3	BMS	Battery management system
4	SOC	State of charge
5	SOH	State of health
6	DM	Direct-measurement model
7	HDM	Hybrid direct-measurement model
8	HECM	Hybrid equivalent-circuit model
9	DD	Data-driven neural model
10	ECM	Equivalent circuit model
11	EKF	Extended Kalman filter
12	MAE	Mean absolute error
13	RMSE	Root-mean-square error
14	ADC	Analog-to-digital converter
15	CV	Constant-voltage phase

16	OCV	Open-circuit voltage
17	GRU	Gated recurrent unit
18	LSTM	Long short-term memory
19	LFP	Lithium iron phosphate
20	<i>Symbols</i>	
21	t	Time
22	$I(t)$	Current signal at time t
23	$U(t)$	Voltage signal at time t
24	$T(t)$	Temperature signal at time t
25	$\widehat{\text{SOC}}(t)$	Estimated state of charge at time t
26	$\widehat{\text{SOH}}(t)$	Estimated state of health at time t
27	$\text{SOC}_{\text{ref}}(t)$	Reference SOC target at time t
28	$\text{SOH}_{\text{ref}}(t)$	Reference SOH target at time t
29	ΔI	Current bias or offset
30	ΔU	Voltage bias or offset
31	ΔT	Temperature bias or offset
32	σ_I	Standard deviation of injected current noise
33	σ_U	Standard deviation of injected voltage noise
34	σ_T	Standard deviation of injected temperature noise
35	Q_c	Online cumulative charge state used by the SOC models
36	C_{nom}	Nominal capacity
37	C_{eff}	Effective capacity used inside the estimator
38	U_{max}	Upper voltage threshold used for full-charge detection
39	U_{tol}	Voltage tolerance used in the full-charge condition
40	U_1, U_2	Polarization voltages of the first and second RC branch
41	ΔMAE	Error increase relative to baseline

42 1. Introduction

43 1.1. Motivation and practical relevance

44 Reliable battery state estimation is a core BMS function because SOC and
45 SOH determine safety margins, usable energy, operating limits, charge con-
46 trol, and degradation-aware system management in both mobile and stationary
47 lithium-ion battery applications [1, 2, 3]. Practical deployment quality is not de-
48 fined by clean-condition accuracy alone, because real BMS operation is exposed
49 to sensor bias, additive noise, initialization uncertainty after restart, missing
50 samples, irregular sampling, communication interruptions, and transient signal
51 spikes caused by measurement-chain limitations or disturbances in the surround-
52 ing system [4, 5, 6]. The relevant engineering question is therefore not only
53 whether an estimator achieves low MAE under nominal measurements, but also
54 whether it converges back to a reliable state after disturbed initialization and
55 whether it remains stable under physically plausible measurement corruption.

56 As summarized conceptually in Figure 1, the present benchmark treats
57 deployment-facing performance as a three-dimensional problem consisting of
58 nominal accuracy, convergence behavior after state inconsistency, and robust-
59 ness against disturbed measurements and signal-integrity faults, embedded within
60 broader BMS requirements such as hardware constraints and implementation
61 effort. The present study therefore isolates the performance branch and com-
62 pares representative estimator classes under identical measurement-side distur-
63 bances and a matched embedded-feasibility constraint. The benchmark shows
64 that strong nominal accuracy does not necessarily imply strong disturbance
65 performance and that the estimator classes respond differently to bias, missing
66 samples, and re-initialization events.

67 1.2. State of the art in battery state estimation

68 The literature commonly groups battery-state estimators into direct-measurement
69 or integration-based approaches, model-based approaches based on equivalent-
70 circuit or electrochemical models, hybrid approaches that combine physical

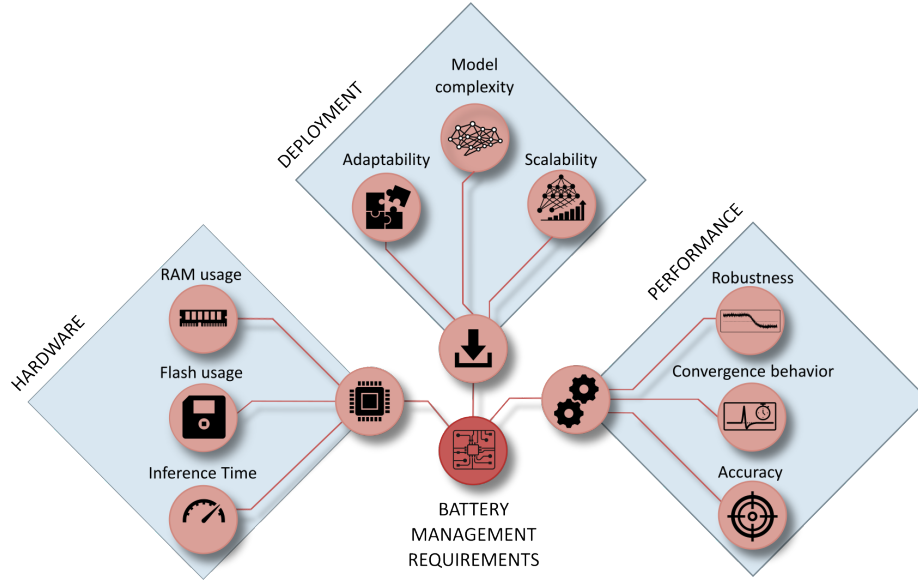


Figure 1: Conceptual overview of deployment-oriented requirements for embedded BMS state estimation. The broader requirement space includes implementation effort, hardware constraints, and estimator performance, whereas the present study focuses on the performance branch and evaluates baseline accuracy, convergence behavior, and robustness under measurement disturbances.

71 structure with learned components, and purely data-driven models based on
 72 neural sequence learning [7, 1, 2, 3]. Direct-measurement approaches, typically
 73 based on Coulomb counting, are attractive because they are simple, computa-
 74 tionally inexpensive, and easy to deploy; however, they remain structurally sen-
 75 sitive to current bias, missing data, and initialization errors because they lack an
 76 internal correction mechanism [7, 4]. Model-based approaches, especially ECM-
 77 and observer-based estimators, offer physical interpretability and voltage-based
 78 correction capability, but their practical behavior depends strongly on the qual-
 79 ity of model parametrization, operating-point observability, and measurement
 80 quality [8, 2, 4].

81 Hybrid estimators seek to combine physical state propagation with learned
 82 adaptation of capacity, parameters, or auxiliary states, which has made joint
 83 SOC/SOH estimation an active research direction [9, 10, 11]. This combination

can improve baseline calibration, while it does not necessarily eliminate structural sensitivity to measurement disturbances. Data-driven approaches based on recurrent or sequence models can exploit nonlinear cross-relations between current, voltage, temperature, and derived online features, and recent work has also demonstrated their embedded feasibility through TinyML-oriented implementations, pruning, and quantization [12, 13, 14, 15, 3].

1.3. Performance and embedded deployment gap

A large share of the SOC estimation literature still emphasizes clean-data accuracy, average error metrics, or computational efficiency, while comparatively fewer studies analyze how estimators behave under realistic measurement faults and timing corruption [1, 13, 14, 15]. Existing robustness studies often remain confined to a single model family, for example observer-based robustness against modeling and measurement errors [4] or robust estimation concepts for specific BMS architectures under noise and uncertainty [5], whereas cross-family comparisons under the same disturbance protocol and under comparable embedded constraints remain limited. Benchmarking studies in the broader BMS context underline the importance of reproducible scenario design and deployment-relevant validation, but they do not by themselves close the gap between clean-data estimator comparison and measurement-level performance benchmarking across fundamentally different estimator classes [6]. This gap is especially relevant in embedded BMS design, where estimator selection must jointly consider nominal accuracy, convergence, robustness under disturbed measurements, and computational constraints.

1.4. Contribution and study objectives

This work defines deployment-facing estimator performance as the joint assessment of nominal accuracy, convergence behavior after state inconsistency, and robustness under realistic disturbances applied only to measurable inputs and their timing. The study compares four deployment-prepared representatives of major estimator families, namely a direct-measurement model (DM), a

113 hybrid direct-measurement model with learned SOH support (HDM), a hybrid
 114 equivalent-circuit model with voltage feedback (HECM), and a data-driven neu-
 115 ral model with rolling-window GRU-based SOC estimation and stateful LSTM-
 116 based online SOH context (DD), under identical causal execution conditions.
 117 The disturbances are deliberately restricted to measurement-level and signal-
 118 integrity scenarios, including current bias, current noise, voltage noise, temper-
 119 ature noise, missing samples, irregular sampling, burst dropouts, voltage spikes,
 120 ADC quantization, and initialization errors, so that the benchmark remains
 121 physically interpretable and experimentally reproducible.

122 The manuscript should therefore be read as a controlled cross-class esti-
 123 mator benchmark and not as a complete system-level BMS validation cam-
 124 paign. Its role is to isolate estimator-side failure modes under matched data,
 125 matched disturbance definitions, matched online execution rules, and matched
 126 embedded-feasibility constraints. The resulting contribution is a measurement-
 127 only benchmark that separates baseline accuracy, convergence, and disturbed-
 128 operation robustness and translates the observed failure modes into decision-
 129 oriented guidance for embedded BMS estimator selection.

130 **2. Methodology**

131 *2.1. Estimator classes under test*

132 Figure 2 summarizes the causal benchmark chain, from measured input sig-
 133 nals and online feature reconstruction to disturbance injection and the final
 134 global and local performance evaluation. The direct-measurement model (DM)
 135 represents the canonical embedded Coulomb-counting approach with fixed nom-
 136 inal capacity and explicit full-charge reset logic, and serves as the low-complexity
 137 reference for integration-driven SOC estimation. All SOH-dependent estima-
 138 tor variants in the benchmark, namely HDM, HECM, and DD, use the same
 139 shared LSTM-based online SOH estimator; the classes differ only in how this
 140 common SOH signal is consumed inside the SOC pipeline. The hybrid direct-
 141 measurement model (HDM) preserves the same Coulomb-counting core but re-

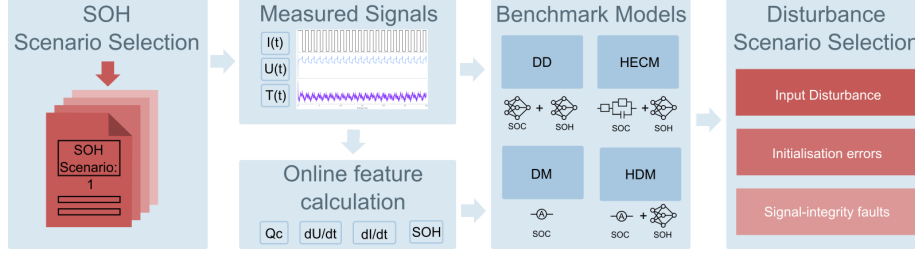


Figure 2: Methodology overview for the causal benchmark chain. Measured battery signals are converted into online auxiliary features, processed by representative estimator classes, exposed to measurement-only disturbances, and finally evaluated through global and local performance metrics.

places the fixed capacity assumption with an online effective capacity derived from this shared SOH estimator, which makes it representative of lightweight hybrid estimation. The hybrid equivalent-circuit model (HECM) represents physics-based state estimation with observer correction, because it combines current-driven state propagation with voltage-residual feedback and SOH-dependent parameterization in a second-order RC equivalent-circuit structure driven by the same shared SOH estimate [8, 2]. The data-driven model (DD) represents a neural estimator pipeline in which SOC is predicted by a lightweight rolling-window GRU sequence model while the shared LSTM-based online SOH estimate is provided as causal context under the same execution constraints [12, 3].

2.2. SOC estimation methods

2.2.1. Direct-measurement model (DM)

The direct-measurement SOC formulation used in DM, HDM, and as an explicit auxiliary feature in DD follows the same online cumulative-charge state, denoted here uniformly as Q_c . It is updated causally from the measured current stream as

$$Q_c(k) = Q_c(k-1) + \frac{I_k \Delta t_k}{3600 \text{ s h}^{-1}}, \quad (1)$$

where I_k is given in ampere and Δt_k in seconds, so that the factor 3600 s h^{-1} converts the increment to ampere-hours. The sign convention is fixed by the

benchmark implementation and the state is reset to $Q_c = 0$ after a sustained constant-voltage full-charge event. In the DM, this same charge state is mapped directly to SOC through the fixed nominal capacity

$$\widehat{\text{SOC}}_{\text{DM}}(k) = \text{clip}\left(1 + \frac{Q_c(k)}{C_{\text{nom}}}, 0, 1\right), \quad (2)$$

154 2.2.2. Hybrid direct-measurement model (HDM)

The HDM preserves exactly the same Coulomb-counting core and differs only in the capacity term. The SOH estimate rescales the usable capacity as

$$C_{\text{eff}}(k) = C_{\text{nom}} \cdot \widehat{\text{SOH}}(k), \quad (3)$$

which yields

$$\widehat{\text{SOC}}_{\text{HDM}}(k) = \text{clip}\left(1 + \frac{Q_c(k)}{C_{\text{eff}}(k)}, 0, 1\right). \quad (4)$$

155 This formulation makes the relation between DM and HDM explicit: both use
 156 the same online charge state Q_c , while only the denominator changes through
 157 online SOH support. In both direct-measurement variants, a sustained near-
 158 maximum voltage condition is interpreted as a constant-voltage full-charge event;
 159 if the voltage remains above $U_{\text{max}} - U_{\text{tol}}$, where U_{tol} denotes the admissible volt-
 160 age margin below the upper threshold U_{max} , for the configured CV duration, the
 161 internal charge state is reset so that $Q_c = 0$ and therefore $\widehat{\text{SOC}} = 1$, which re-
 162 flects the practical assumption that a completed CV phase provides a physically
 163 anchored full-charge reference.

164 2.2.3. Hybrid equivalent-circuit model (HECM)

The hybrid ECM model is implemented as a second-order RC model with the state vector

$$\mathbf{x}_k = \begin{bmatrix} \widehat{\text{SOC}}_k & U_{1,k} & U_{2,k} \end{bmatrix}^\top, \quad (5)$$

where U_1 and U_2 denote the polarization voltages of the two RC branches. The prediction step propagates the state according to

$$\mathbf{x}_k^- = \mathbf{A}_k \mathbf{x}_{k-1} + \mathbf{b}_k I_{k-1}, \quad (6)$$

with

$$\mathbf{A}_k = \text{diag}\left(1, e^{-\Delta t_k/\tau_1}, e^{-\Delta t_k/\tau_2}\right), \quad \mathbf{b}_k = \begin{bmatrix} \eta\Delta t_k/C_b \\ R_1(1 - e^{-\Delta t_k/\tau_1}) \\ R_2(1 - e^{-\Delta t_k/\tau_2}) \end{bmatrix}, \quad (7)$$

where $C_b = C_0\widehat{\text{SOH}}$ is the SOH-scaled available capacity. The corrected terminal-voltage estimate is written as

$$\hat{U}_k = \text{OCV}(\widehat{\text{SOC}}_k, \widehat{\text{SOH}}_k, I_k) + U_{1,k} + U_{2,k} + R_i I_k, \quad (8)$$

and the Kalman update uses the residual between $\hat{U}_k$ and the measured terminal voltage to correct the internal state. In the present implementation, the state propagation is driven by the previous current sample I_{k-1} , whereas the measurement equation uses the current sample I_k available at the correction step. The ECM parameters are not constant. The implementation interpolates R_i , R_1 , R_2 , τ_1 , τ_2 , OCV, and dOCV/dSOC from a preidentified lookup table over SOC and SOH, with separate charge and discharge parameter surfaces. The HECM is implemented as a two-RC observer with SOH-dependent parameter scheduling.

2.2.4. Data-driven SOC model (DD)

The data-driven SOC model is a GRU-based recurrent estimator with the feature vector $\{U \text{ [V]}, I \text{ [A]}, T \text{ [}^\circ\text{C]}, \text{SOH}, Q_c, dU/dt, dI/dt, \Delta t\}$. The benchmarked structurally pruned and fine-tuned checkpoint consists of a single-layer GRU with hidden size 67 followed by an MLP head with one hidden layer of size 96 and a sigmoid output layer. During optimization, the recurrent core is exposed to causal sequence segments of length $L_{\text{SOC}} = 2024$ samples. Within this feature set, Q_c is exactly the same online cumulative-charge state defined in (1). It is included explicitly so that the neural estimator receives the same charge-throughput information that also drives DM and HDM. The derivative channels are computed online as finite differences with respect to the current sampling interval,

$$\frac{dI}{dt}(k) = \frac{I_k - I_{k-1}}{\Delta t_k}, \quad \frac{dU}{dt}(k) = \frac{U_k - U_{k-1}}{\Delta t_k}, \quad (9)$$

and provide the neural estimator with local transient information on current and voltage dynamics that is not directly captured by the absolute channel values alone. The additional feature Δt_k is passed explicitly so that the network can distinguish amplitude changes from sampling-interval changes under irregular timing. During the primary benchmark, each output is calculated from the latest causal 2024-sample window and recurrent state is reset between windows, matching training and validation. No future information is used. Continuous hidden-state forwarding is evaluated only as a separate deployment ablation because it changes the learned inference semantics.

2.3. SOH estimation method

The shared SOH estimator used in the current benchmark is an LSTM-based sequence-to-sequence model that operates on hourly aggregated online features derived from $\{U \text{ [V]}, I \text{ [A]}, T \text{ [}^\circ\text{C]}, \text{EFC}, Q_c\}$, where the aggregation uses the statistics mean, standard deviation, minimum, and maximum for each feature [16, 17, 18, 19]. This transforms the five base channels into a 20-dimensional hourly feature vector. Architecturally, the SOH network first projects the aggregated input through a two-layer feature embedding block with size 128, layer normalization, GELU activations, and dropout, then processes the sequence with a two-layer unidirectional LSTM of hidden size 112, and finally maps the recurrent output through three residual MLP blocks and a prediction head with hidden size 160 to obtain the SOH estimate for each hourly step.

During optimization, the SOH estimator is exposed to causal hourly sequence segments, and benchmark execution processes one hourly aggregate at a time with causal hidden- and cell-state forwarding. This allows the estimator to accumulate degradation context without access to future information; the first hourly output is anchored with the configured initial SOH value before subsequent hourly predictions are taken directly from the recurrent model output. The SOH prediction is updated at hourly cadence and then held piecewise constant between update points, which enforces an online deployment constraint and avoids sample-wise access to future information while still allowing the faster SOC es-

timators to use the latest SOH context. In the HDM, the predicted SOH enters the SOC computation through the effective capacity relation $C_{\text{eff}} = C_{\text{nom}} \cdot \widehat{\text{SOH}}$, so the method remains structurally close to direct measurement while replacing the fixed-capacity assumption. In the HECM, the predicted SOH affects the capacity scaling and the lookup-based ECM parametrization, while in the DD model it is provided as an explicit SOC input feature that contextualizes the current measurement trajectory.

2.4. Model and target-hardware preparation

The software benchmark uses the frozen floating-point checkpoints and scalers selected before holdout evaluation. The compact SOC-GRU has hidden size 67; the shared SOH-LSTM has hidden size 112. Configuration, checkpoint, and scaler hashes are stored with the campaign manifest, and the runner rejects cells outside the declared holdout split unless an explicit diagnostic override is supplied.

The target-hardware experiment evaluates the isolated SOC cores on a NUCLEO-H753ZI board with a 480 MHz STM32H753 Cortex-M7. To separate SOC implementation cost from the shared SOH service, the board receives the precomputed causal SOH trace used by the software reference; the SOH-LSTM itself is not executed on the microcontroller. The hardware results therefore measure numerical equivalence, latency, and memory of the SOC implementations rather than end-to-end SOC-SOH system latency. Device time is measured per valid inference with the data-watchpoint-and-trace cycle counter. Each cell sequence is replayed three times, firmware section sizes are extracted from the release ELF images, and predictions are compared with both the software implementation and dataset SOC. Reported RAM is static ELF allocation including the linker-reserved heap/stack block, not measured peak stack or dynamic activation memory.

2.5. Performance benchmark design

The benchmark follows a strict measurement-only principle: only quantities that a real BMS measures or derives online from measured data are manipu-

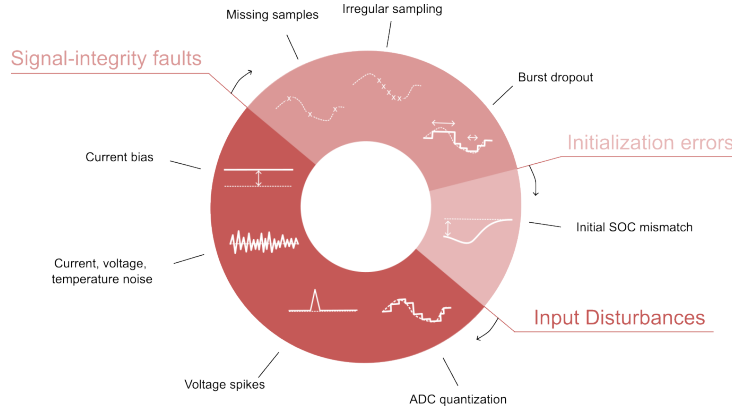


Figure 3: Taxonomy of the disturbance scenarios used in the performance benchmark, grouped into input disturbances, initialization errors, and signal-integrity faults.

lated, namely current, voltage, temperature, sample availability, and sampling
time, whereas artificial parameter mismatch and synthetic hidden-state corrup-
tion are deliberately excluded [6]. Figure 3 groups the benchmark into input
disturbances, initialization errors, and signal-integrity faults, while Table 1 de-
fines the exact disturbance implementations used in the study.

The disturbance magnitudes are not intended as one-to-one copies of a single
field specification. Instead, they were selected from three justification classes:
deployment-level sensing limitations, adverse measurement corruption, and de-
liberate signal-integrity stress tests [20, 21, 22, 23, 24, 25]. The main benchmark
matrix follows the scenario families shown in Figure 3 and provides the basis
of the main result figures and tables. ADC quantization is part of the same
scenario set and is reported in Figure 16. Overall, the benchmark is designed
as a controlled and reproducible scenario study that exposes estimator-specific
failure modes under a common disturbance protocol rather than emulating the
complete system stack of a production BMS.

Table 1: Disturbance configurations used in the benchmark and their practical rationale. Exact numeric values are fixed benchmark settings; the cited sources motivate disturbance type and magnitude range.

Scenario	Benchmark setup	Practical origin / rationale
ADC quantization	$\Delta I = 0.01 \text{ A}$; $\Delta U = 0.005 \text{ V}$; $\Delta T = 0.5 \text{ }^\circ\text{C}$	ADC steps; sensor scaling; rounding; coarse deployment-level discretization [20, 21, 26]
Current gain error	Signed gain sweep: $I_{\text{meas}} = I(1 + b)$ with $b = \pm 0.5\%$, $\pm 1.5\%$, and $\pm 3.0\%$	Sensor gain error; miscalibration; drift; upper bound as stress case [21, 22, 23]
Current noise	Gaussian; $\sigma_I = 0.02 \text{ A}$ and 0.10 A	Sensing-chain noise; EMI pickup; low and high sensor-noise levels [4, 5]
Voltage noise	Gaussian; $\sigma_U = 0.01 \text{ V}$	Wiring noise; front-end noise; poor filtering; EMI [4, 20, 24]
Temperature noise	Gaussian; $\sigma_T = 1.0 \text{ }^\circ\text{C}$	Sensor tolerance; conversion noise; thermal gradients; sensor-to-cell mismatch [26, 11]
Initial SOC error	Initial mismatch: -0.10 SOC ; DD via perturbed online Q_c	Restart; storage; missing recent history; weak LFP OCV anchoring [4, 8]
Missing samples	Every 50th sample frozen; separate random 2% loss	Packet loss; blocked updates; logger dropouts; periodic and random data freeze [5, 25]
Irregular sampling	Uniform jitter: $\pm 0.1 \text{ s}$, $\pm 0.5 \text{ s}$, and $\pm 0.9 \text{ s}$ around nominal 1 s grid	Scheduling jitter; bus latency; asynchronous acquisition [6, 5, 25]
Burst dropout	One central frozen gap: 3600 s	Communication interruption; frozen task; stuck data stream; severe recovery case [6, 5]
Voltage spikes	Single-sample spikes: $\pm 0.20 \text{ V}$ every 1000 samples (1000 s)	EMI; switching disturbance; corrupted voltage samples [24]

2.6. Metrics and evaluation criteria

For deployment-oriented interpretation, the benchmark results are synthesized along three performance dimensions: nominal accuracy under baseline conditions, convergence behavior after initialization mismatch or interrupted signal continuity, and robustness under sustained or transient measurement corruption. Disturbance-related global performance is quantified through MAE, RMSE, bias, maximum error, and the 95th percentile absolute error, because these metrics provide a compact view of average performance, systematic deviation, and tail behavior across the selected evaluation windows. Local disturbance performance is analyzed through scenario-specific metrics such as jump

counts, output variance, absolute-error variance, and excess rolling MAE, because several disturbances do not primarily manifest as window-wide MAE collapse but as short-term output fluctuations, delayed drift, or slow re-synchronization. Whenever a disturbance mask exists, the analysis further separates pre-disturbance, disturbed, and post-disturbance behavior in order to quantify disturbance-window error growth, recovery time, and residual post-disturbance error.

ADC quantization, current bias, voltage noise, temperature noise, missing samples, and irregular sampling are interpreted primarily through global metrics, whereas burst dropout, voltage spikes, current-noise detail, and initialization mismatch additionally use dedicated local evidence because their most relevant effects are episodic or recovery-related. Recovery comparisons use one estimator-independent criterion: absolute SOC error below 0.02 for a continuous 300 s, evaluated up to a common 24 h censoring horizon. Runs that do not recover are retained as censored observations rather than silently omitted. The comparison reports recovery-or-censoring time, the censored fraction, initial post-event error, MAE after 1 h and 6 h, and excess-error area under the curve. The requested -0.10 initialization mismatch is mapped into each estimator’s available online state: the coulomb-counting SOC initialization for DM and HDM, the EKF SOC state for HECM, and the capacity-equivalent Q_c offset for DD. Because the DD mapping passes through the recurrent network rather than directly fixing its output, the realized initial output error is reported for every model before recovery is compared.

The independent statistical unit is the holdout cell, not an individual time sample or selected window. Predeclared SOH windows are first averaged within each cell and random seed; the six cells then receive equal macro weight. Hierarchical bootstrap resampling with seeds nested inside cells and 10,000 repetitions provides 95% confidence intervals. Scenario effects are evaluated as paired, cell-matched differences from baseline, and estimator pairs are compared using exact two-sided sign-flip tests, raw mean and median differences, paired standardized effect sizes, and Holm-adjusted p -values. Because only six independent cells are available, effect magnitudes and confidence intervals are treated as primary

291 evidence and p -values as supporting evidence.

292 3. Experimental Setup

293 3.1. Dataset and data acquisition

294 The benchmark is based on the MGFarm LFP 18650 dataset, which was
295 generated in a laboratory environment as a controlled design-of-experiments
296 study on cyclic aging and operating conditions and combines long full-cell tra-
297 jectories with high-resolution measured signals and repeated checkup structures
298 [27, 28, 29]. Model development and final testing use a fixed cell-disjoint split.
299 SOC and SOH training use C01, C03, C05, C11, C17, and C23; validation
300 uses C07, C19, and C21; and the independent holdout benchmark uses C09,
301 C13, C15, C25, C27, and C29. The six test cells span distinct aging, ther-
302 mal, duration, and load envelopes, as summarized in Figure A.21. Conclusions
303 are therefore limited to the available LFP chemistry and measured operating
304 envelope; the study does not claim cross-chemistry or pack-level generalization.

305 The holdout cells are assigned to descriptive load groups before model re-
306 sults are inspected, using visible separation in the measured 95th-percentile
307 absolute C-rate: C25 and C27 form the low-load group, C09, C13, and C15 the
308 middle-load group, and C29 the high-load case. Because the high-load group
309 contains one cell, its group-by-SOH results are explicitly exploratory and are
310 not interpreted as population-level high-load inference. Within each trajectory,
311 performance is additionally stratified by reference SOH (fresh: $\text{SOH} \geq 0.90$;
312 mid-life: $0.80 \leq \text{SOH} < 0.90$; aged: $\text{SOH} < 0.80$), instantaneous absolute C-
313 rate (< 0.5 , $0.5\text{--}1.5$, and ≥ 1.5), temperature ($\leq 30^\circ\text{C}$, $30\text{--}35^\circ\text{C}$, and $> 35^\circ\text{C}$),
314 and SOC (< 0.20 , $0.20\text{--}0.80$, and > 0.80). Absent states are reported as not
315 available and are never imputed.

316 3.2. Reference targets and preprocessing

317 The reference preprocessing first computes the signed transferred charge as
318 $Q_k = I_k \Delta t_k / 3600$ and the absolute transferred charge as $|Q_k|$, from which the

319 capacity during dedicated capacity-test intervals is obtained as $C_{\text{ref}} = \sum_k |Q_k|$.
 320 The reference SOH target is then defined offline as $\text{SOH}_{\text{ref}} = C_{\text{ref}}/C_{\text{ref},0}$, where
 321 $C_{\text{ref},0}$ denotes the first measured reference capacity of the trajectory, and the
 322 resulting capacity and SOH columns are interpolated between the dedicated
 323 capacity-test anchors. The reference SOC target used in the benchmark corre-
 324 sponds to the offline Coulomb-counting target $\text{SOC}_{\text{ref}} = 1 + Q_c/C_{\text{ref},0}$, where
 325 the cumulative charge state Q_c is propagated from the signed charge and reset
 326 to 0 at the upper voltage anchor and to $-C_{\text{ref},0}$ at the lower voltage anchor dur-
 327 ing preprocessing. These reference targets are intentionally treated as offline
 328 evaluation quantities, because they rely on dataset-side processing and anchor
 329 logic that are not fully available to an online estimator during deployment.

330 *3.3. Test trajectory and benchmark scenarios*

331 All estimators are evaluated on every available holdout-cell/SOH combina-
 332 tion under the same disturbance protocol. One representative 24-h window is
 333 frozen for each available fresh, mid-life, and aged state, yielding 16 windows;
 334 C27 contributes only a fresh window because its measured SOH does not fall
 335 below 0.90. Windows start at a measured full-charge anchor ($\text{SOC} \geq 0.98$,
 336 $U \geq 3.58 \text{ V}$) and are selected as measured-feature medoids without using es-
 337 timator outputs or errors. The one-hour dropout uses 48 h from the same
 338 anchor to retain approximately 24 h of post-gap observation. The shared SOH
 339 LSTM receives 192 h of undisturbed causal context before every scored window,
 340 matching its trained sequence length. Figure A.23 summarizes this protocol.

341 The 19 predeclared cases include the nominal baseline, two current-noise
 342 levels, voltage and temperature noise, three current-bias levels, voltage and
 343 temperature offsets, ADC quantization, initial SOC mismatch, periodic and
 344 random sample loss, three timing-jitter levels, a one-hour burst dropout, and
 345 randomized-sign voltage spikes. Gaussian sensor-noise cases use 10 determinis-
 346 tic seeds; random sample loss, jitter, and spike cases use 5 seeds; deterministic
 347 cases run once. Figure A.22 maps each case to the manipulated measurement
 348 or state and marks the selected paired reference-SOH ablations. Input distur-

bances are applied directly to measured channels, initialization errors only where structurally equivalent, and signal-integrity scenarios to sample availability or timing while preserving causal estimator execution.

3.4. Implementation details

Every estimator is executed causally, and all auxiliary quantities required by the models are rebuilt directly from disturbed inputs. This applies to Q_c , EFC, dU/dt , dI/dt , and hourly SOH aggregates, so no hidden dataset-side feature leaks into the estimator input. The SOH LSTM forwards hidden and cell states across hourly updates after its 192-h initialization context. The primary SOC-GRU evaluation preserves its trained sequence-to-one semantics: each prediction uses the latest 2024-sample rolling window with recurrent state reset between windows. A faster continuous-stateful stream is treated only as a deployment ablation because it changed full-trajectory DD accuracy and therefore cannot silently replace the trained benchmark semantics. In freeze and dropout scenarios, disturbed or frozen measurements propagate consistently into derived online features. Direct-measurement estimators use the same CV-triggered reset that defines the full-charge start, whereas the DD initial-SOC test uses an equivalent offset of online Q_c because the network exposes no explicit SOC state. The environment enforces identical capacity assumptions, disturbed streams, targets, and metrics across all estimators.

4. Results

4.1. Baseline performance

Figure 4 summarizes nominal performance across the 16 frozen windows. Equal weighting of the six holdout cells yields a different ranking from the previous single-trajectory analysis: DD has the lowest baseline MAE at 0.0258 (95% CI 0.0189–0.0323), followed by HECM at 0.0335 (0.0246–0.0429), HDM at 0.0405 (0.0297–0.0506), and DM at 0.0678 (0.0491–0.0809). The RMSE ranking is identical, with values of 0.0300, 0.0389, 0.0438, and 0.0733, respectively. Cell-level points reveal substantial between-cell variation, particularly for DM, so

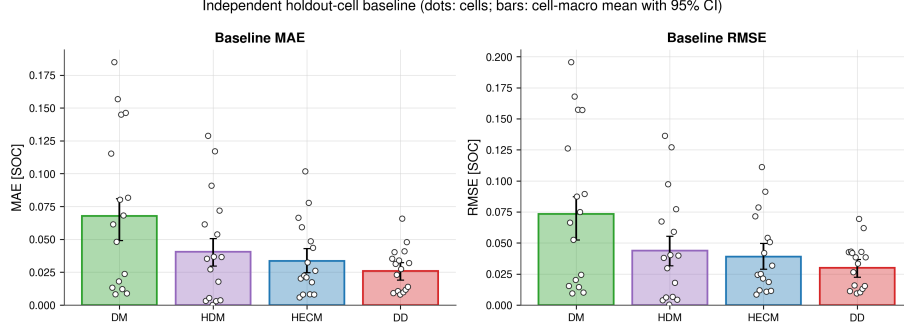


Figure 4: Baseline performance across six independent holdout cells. Dots show cell/SOH-window values; bars and error bars show the equal-weight cell-macro mean and hierarchical 95% confidence interval. DD has the lowest macro MAE and RMSE, while DM has the largest error and between-cell spread.

the bars must be interpreted as cell-macro summaries rather than performance on a homogeneous trajectory.

4.2. Current-bias sensitivity

Figure 5 summarizes the adverse response to a signed current-gain sweep at $\pm 0.5\%$, $\pm 1.5\%$, and $\pm 3.0\%$. For each magnitude and holdout cell, the larger MAE change from the two signs is reported because the opposite sign can partially compensate a pre-existing baseline error. The resulting worst-case degradation increases monotonically with gain-error magnitude for every estimator. DM and HDM show the strongest response, HECM is less sensitive, and DD is least sensitive. At 3%, the cell-macro MAE increases by 0.0107 for DM, 0.0100 for HDM, 0.0059 for HECM, and 0.0038 for DD.

The continuous C29 lifecycle analysis in Figure 6 separates three mechanisms that independent 24-h windows cannot resolve. The bias contribution varies over life because its signed Coulomb-counting error interacts with charge direction and operating state. Between full charges, the penalty generally grows within an interval, but full-charge anchoring can partially reduce or restructure it. The process is therefore neither an unrestricted monotonic drift nor a complete reset. Across the full replay, DD retains the smallest adverse penalty, whereas DM and

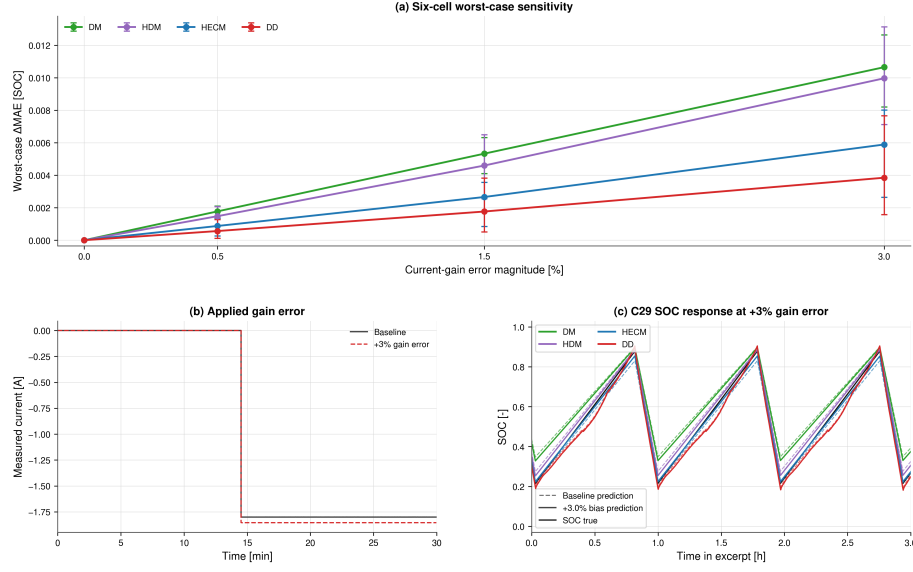


Figure 5: Current-gain sensitivity. (a) Six-cell adverse-direction ΔMAE for each matched $\pm b$ pair; error bars are hierarchical 95% confidence intervals. (b) Representative current before and after the +3% gain error. (c) C29 SOC trajectories over three charge–discharge intervals. The adverse-pair summary prevents signed compensation of pre-existing baseline bias from being misread as improved robustness.

396 HDM remain most exposed to accumulated current error.

397 4.3. Noise sensitivity

398 Repeated sensor noise is weaker than persistent current gain error, but the
 399 response depends on channel and estimator structure. At $\sigma_I = 0.10$ A, HECM
 400 has the largest macro penalty ($\Delta\text{MAE} = 0.0056$), whereas DM changes by
 401 -0.0009 and HDM and DD remain near zero. Small negative changes are paired
 402 finite-window effects, not evidence that noise improves an estimator in general.
 403 Voltage noise ($\sigma_U = 0.01$ V) produces a 0.0014 penalty for DM and 0.0013
 404 for HDM, while HECM and DD remain below 0.0003. Temperature noise is
 405 neutral for DM and small for the SOH-aware models. Figure 7 shows the local
 406 transmission of current noise, while Figure 8 reports repeated six-cell effects
 407 with hierarchical uncertainty.

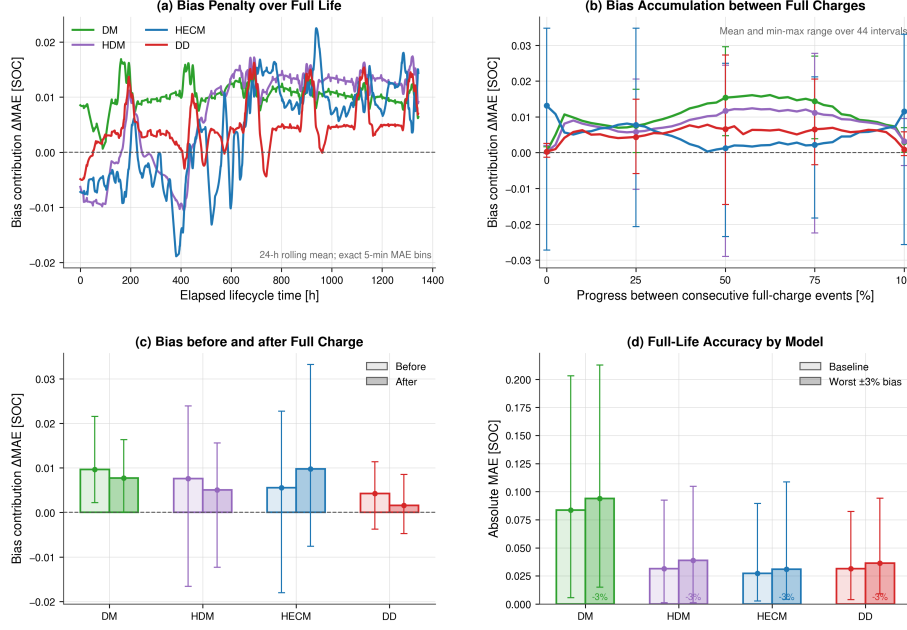


Figure 6: Current-bias accumulation over the continuous C29 life replay. (a) Rolling 24-h bias contribution over elapsed life. (b) Mean and range over normalized intervals between consecutive full-charge events. (c) Penalty immediately before and after full-charge events. (d) Full-life baseline and adverse-bias MAE. Full-charge events reduce or restructure accumulated error but do not erase the lifetime penalty uniformly.

4.4. Initialization error and recovery

Initialization performance is evaluated using one model-independent target criterion: the shifted output must remain within 0.02 SOC of the paired correct-initialization output for at least 300 s, with censoring at 24 h. The same requested -0.10 SOC-equivalent mismatch is mapped to the explicit DM/HDM Coulomb-counting state, the HECM EKF SOC state, and a Q_c offset of $-0.10C_{\text{nom}}$ for DD. Figure 9 reports the realized shift rather than assuming that these internal mappings create identical outputs. Across cells, HECM recovers fastest at 1.03 h on average, followed by DD at 5.57 h; DM and HDM both average 22.28 h and are frequently close to the censor horizon. Thus, observer correction

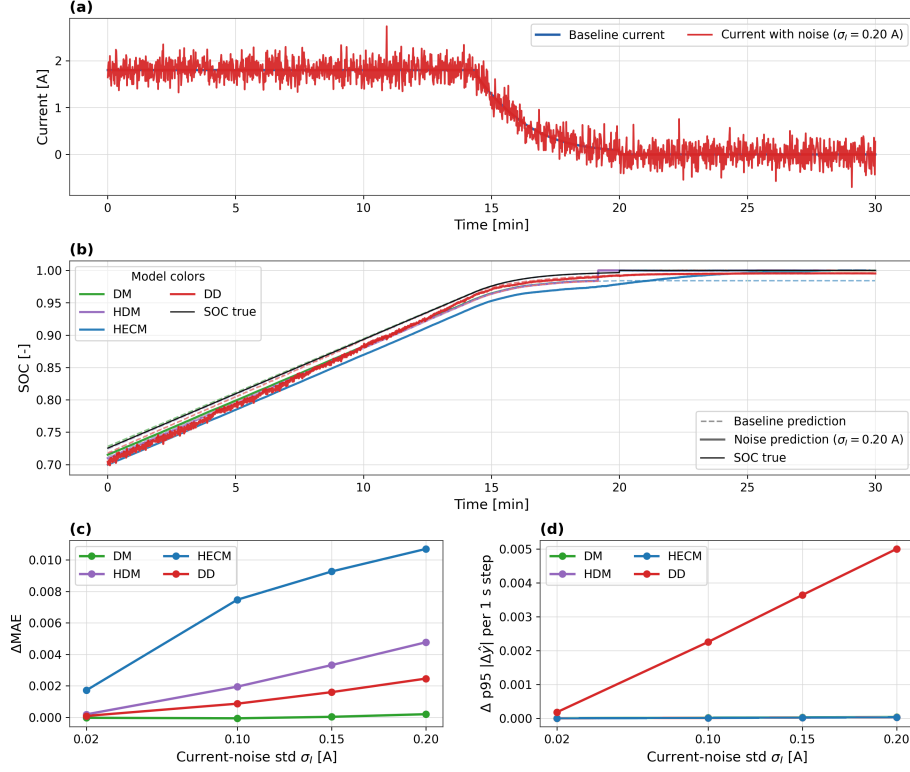


Figure 7: Noise sensitivity under additive current noise. The figure combines an applied current-noise example, a matched SOC comparison in the same 30 min window, the global MAE penalty, and the local output-variability sensitivity across noise levels.

is most effective for the imposed initialization mismatch, whereas integration-based states generally require a later physical anchor.

4.5. Missing samples and signal integrity

Sample loss is substantially more consequential than timing jitter. Periodic freezing of every 50th sample increases MAE by 0.0071, 0.0064, 0.0028, and 0.0011 for DM, HDM, HECM, and DD, respectively; random 2% loss produces corresponding increases of 0.0079, 0.0080, 0.0038, and 0.0026. By contrast, even the ± 0.9 s jitter case remains within approximately 6×10^{-4} SOC of baseline for every model, with confidence intervals generally spanning zero. DD therefore

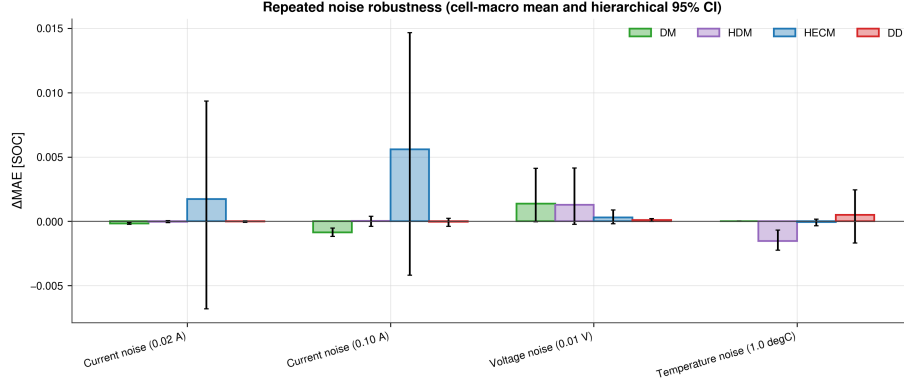


Figure 8: Repeated noise robustness across six holdout cells. Bars show cell-macro ΔMAE and error bars show hierarchical 95% confidence intervals with random seeds nested within cells. HECM is most affected by high current noise, while most other channel-model combinations remain close to baseline relative to the between-cell uncertainty.

has the smallest sample-loss penalty, while DM and HDM retain their structural dependence on uninterrupted charge integration.

The one-hour burst dropout is a distinct recovery problem. Its macro ΔMAE is largest for HECM (0.0221), followed by HDM (0.0134), DD (0.0134), and DM (0.0067). After measurements resume, DD has the shortest mean recovery-or-censor time (4.13 h), then DM (5.13 h), HDM (5.67 h), and HECM (12.45 h). The representative trace in Figure 11 shows why a global score is insufficient: frozen outputs separate strongly from the changing reference during the gap, and model-specific re-synchronization begins only after fresh measurements return.

4.6. Spike sensitivity

Single voltage spikes demonstrate why local and global metrics must be reported together. Full-window ΔMAE remains below 9×10^{-4} SOC for every estimator and is effectively zero for DM, HDM, and DD. Event-aligned analysis, however, identifies a reproducible DD transient: its mean spike-sample penalty across cells is approximately 7.7×10^{-3} SOC, compared with near-zero direct penalties for the other classes. Figures 12 and 13 further show that the DD response varies across cells and SOH states and then decays after the event. This

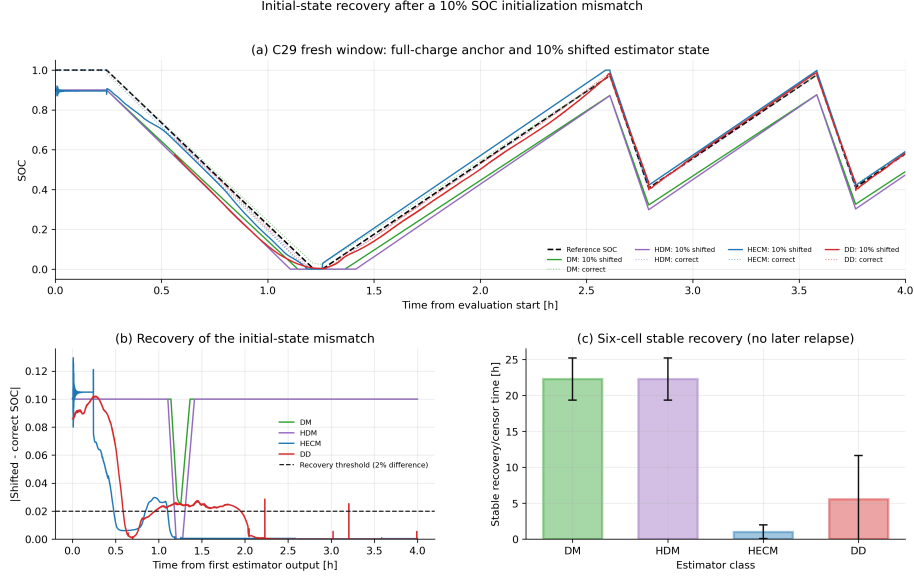


Figure 9: Recovery after a 10% SOC-equivalent initialization mismatch. (a) Representative paired trajectories. (b) Difference between shifted and correctly initialized outputs with the common 2% threshold. (c) Stable recovery-or-censor time across the six holdout cells with hierarchical 95% confidence intervals.

pattern is consistent with a one-sample voltage impulse entering both the absolute voltage and dU/dt channels and perturbing the rolling recurrent prediction. It is a mechanism-supported interpretation, not proof of a unique causal input channel because no channel-removal ablation was performed.

4.7. Shared-SOH ablation

The primary comparison intentionally supplies one common causal LSTM-SOH trace to HDM, HECM, and DD. A paired ablation repeats selected scenarios with dataset reference SOH to quantify this shared-service confounder. Reference SOH reduces baseline MAE by 0.0363 for HDM and 0.0164 for HECM, but increases DD MAE by 0.0049 on average. The same direction persists for most noise, sample-loss, and dropout cases. At 3% current gain error, reference SOH still improves HDM and HECM but worsens DD by 0.0091. This does not imply that a less accurate SOH estimate is intrinsically beneficial; rather, DD

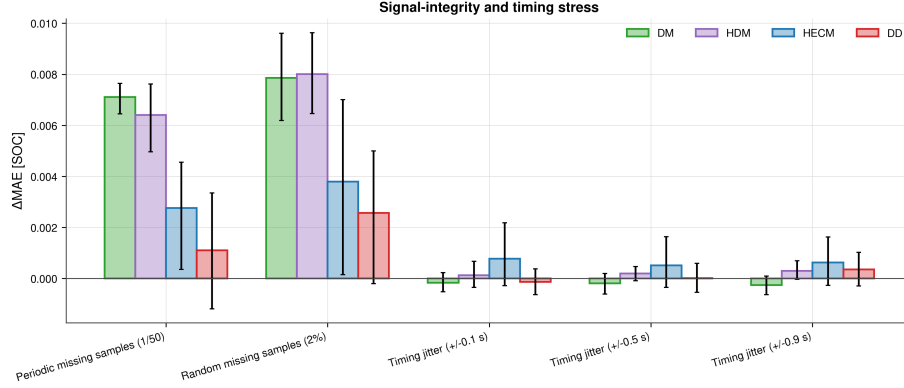


Figure 10: Six-cell signal-integrity and timing stress. Bars show cell-macro ΔMAE and hierarchical 95% confidence intervals for periodic and random sample loss and three timing-jitter levels. Sample loss clearly dominates timing jitter under the tested protocol.

and the hybrid models consume SOH differently and were calibrated with different feature interactions. The LSTM-SOH primary results therefore represent deployable pipeline performance, while the reference-SOH runs are explanatory ablations rather than replacement rankings.

4.8. Cross-scenario comparison

Figure 14 condenses the complete measurement-only campaign. Persistent and missing-information faults dominate the global penalties: current-gain error rises with magnitude for all models, periodic/random sample loss particularly affects DM and HDM, and the one-hour dropout produces the largest penalty for HECM. By contrast, the three timing-jitter levels and voltage spikes remain near zero in full-window MAE, although the spike analysis above reveals a DD-local transient. DD has the smallest adverse current-bias penalty and lowest sample-loss penalties; HECM has the largest high-current-noise and dropout penalties but the fastest initialization recovery. The final data therefore favor DD in the aggregate robustness score without implying uniform dominance in every individual mechanism.

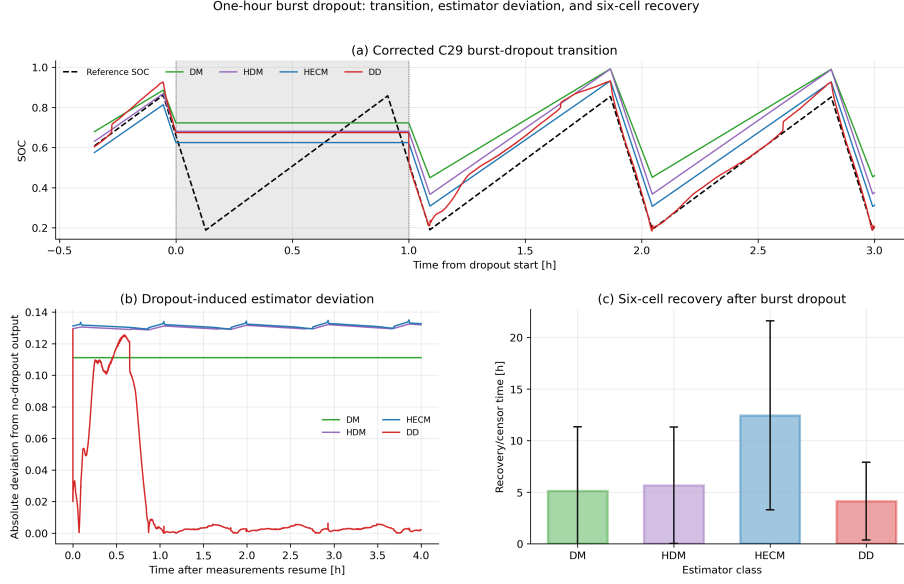


Figure 11: One-hour burst-dropout transition. (a) Baseline outputs (dashed) and dropout outputs (solid) around the shaded missing-input interval. (b) Absolute deviation from the paired no-dropout output after measurements resume. (c) Six-cell recovery-or-censor time under the common 2%/300-s criterion.

4.9. ADC quantization

The tested ADC steps ($\Delta I = 0.01$ A, $\Delta U = 0.005$ V, and $\Delta T = 0.5$ °C) preserve the local signal shape but affect the estimator classes differently. DM and HDM increase by 0.0037 and 0.0034 MAE, respectively. HECM changes by -0.0018 and DD by $+0.0002$; both confidence intervals overlap zero. Thus, quantization at these deliberately coarse steps is secondary to current gain error, sample loss, and burst dropout in this dataset, but it is not negligible for the integration-based models.

4.10. Target-hardware validation

The STM32 experiment first verifies numerical equivalence before comparing speed or memory. Across the six nominal cell sequences, mean hardware–software MAE remains below 4×10^{-4} percentage points for every SOC core and reaches 3.41×10^{-6} percentage points for DD (Figure 17). These differences

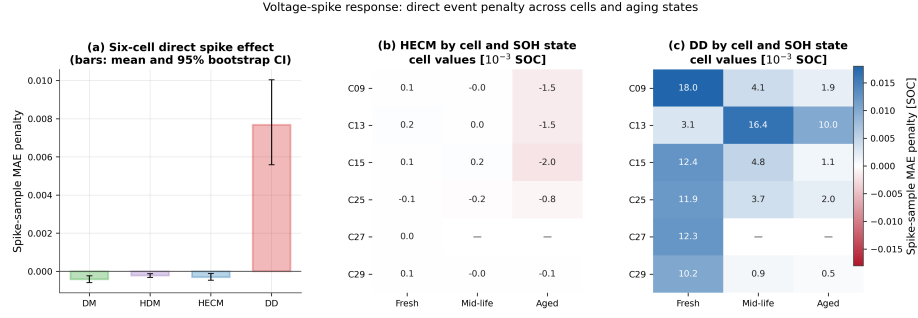


Figure 12: Voltage-spike sensitivity across six cells. (a) Mean event-sample MAE penalty with hierarchical 95% confidence intervals. (b,c) Cell- and SOH-state-resolved penalties for HECM and DD, respectively; absent cell/state combinations are left blank. The event-aligned metric exposes the local DD response that is obscured in full-window Δ MAE.

are orders of magnitude below the estimator-to-dataset errors and show that the float32 C implementations reproduce their software references sufficiently closely for deployment measurements.

The isolated DM and HDM SOC updates both have sub-microsecond median latency (0.171 and 0.165 μ s), HECM requires 23.9 μ s, and continuous-state DD requires 424.9 μ s (Figure 18). All are below the nominal one-second sample interval. The DD value in this cross-model latency comparison is the continuous-state deployment variant; exact 2024-sample rolling-window semantics are evaluated separately because recomputing the full sequence at every sample is much slower.

Release images occupy 27.0, 27.1, 470.1, and 116.1 KiB flash for DM, HDM, HECM, and DD, respectively. Static RAM is approximately 4.8 KiB for DM/HDM, 4.9 KiB for HECM, and 69.8 KiB for rolling-window DD (Figure 19). HECM flash is dominated by its parameter surfaces, while the DD rolling window dominates static RAM. These values describe the isolated firmware images and do not include the shared SOH-LSTM service; static RAM also remains a lower bound on peak runtime memory.

Figure 20 makes the DD deployment trade-off explicit. Exact rolling-window inference preserves the trained software semantics but requires roughly 724 ms

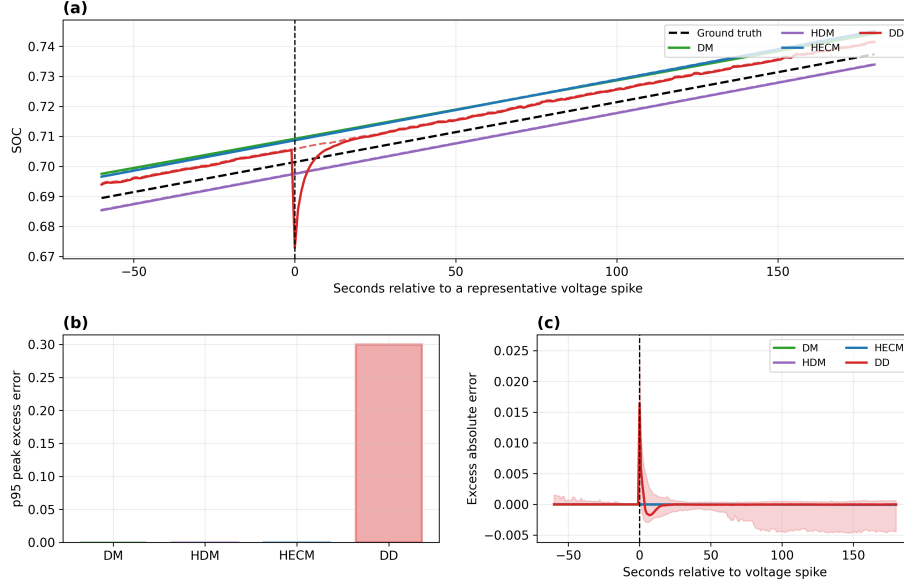


Figure 13: Event-aligned voltage-spike response. (a) Representative SOC trajectories around the injected spike; the dashed black line is reference SOC. (b) Model-wise 95th percentile peak excess error. (c) Mean excess absolute error aligned across repeated spikes, with the DD uncertainty envelope. The local response decays rapidly and is therefore weakly represented by full-window MAE.

505 median inference time and 69.8 KiB static RAM. Continuous-state inference
 506 reduces latency to approximately 0.425 ms and RAM to 6.5 KiB while retaining
 507 similar error on the six nominal sequences. Periodically resetting that state has
 508 similar resource use but creates much larger transient errors, with a maximum
 509 SOC error near 28 percentage points. Continuous state is therefore the practical
 510 hardware candidate, but it remains an inference-semantics ablation rather than
 511 a drop-in replacement for the primary rolling-window software benchmark.

512 5. Discussion

513 5.1. Model-specific performance behavior

514 DM behaves as expected for a pure integration estimator: it has the largest
 515 nominal error and is sensitive to persistent gain error and missing samples be-

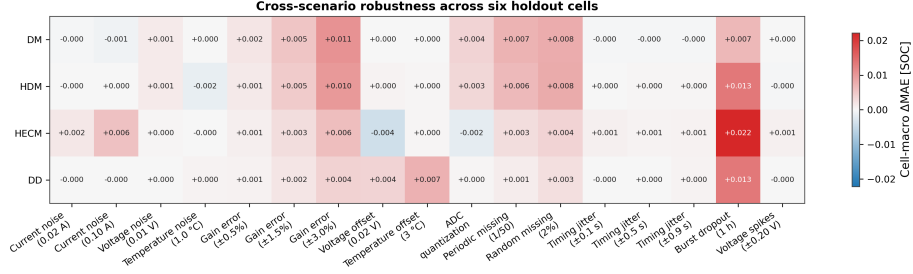


Figure 14: Cell-macro Δ MAE across the complete six-cell measurement-only campaign. Positive values denote degradation relative to each model’s paired baseline; small negative values reflect finite-window signed effects and are not interpreted as beneficial disturbances.

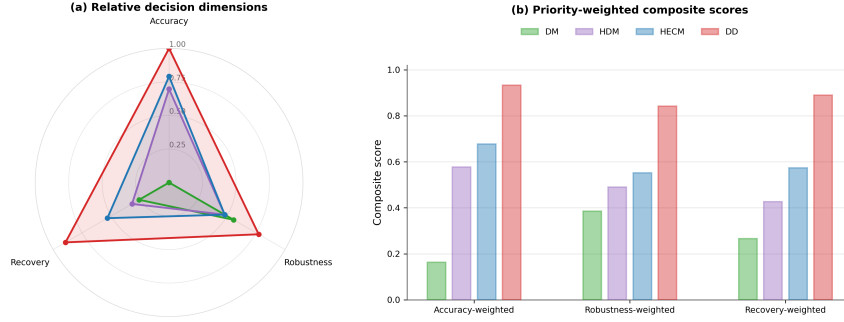


Figure 15: Decision-oriented synthesis. (a) Relative min-max-normalized accuracy, robustness, and recovery dimensions. (b) Composite scores under three illustrative 60/20/20 priority profiles. Scores summarize the present benchmark only and do not replace scenario-level evidence.

cause no voltage-feedback correction is available. Full-charge anchoring limits unbounded drift, but the lifecycle experiment shows that it does not erase the accumulated penalty uniformly. HDM changes the capacity denominator through SOH and therefore reduces some aging-related mismatch, yet it retains the same integration mechanism. Its similar current-bias and sample-loss response is consequently structural rather than incidental.

HECM combines current-driven state propagation with a voltage residual

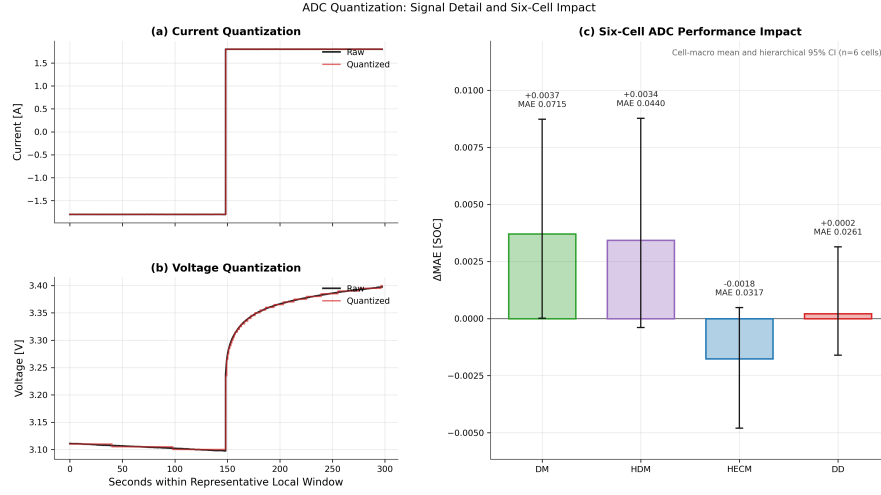


Figure 16: ADC quantization. (a,b) Representative raw and quantized current and voltage. (c) Six-cell Δ MAE with hierarchical 95% confidence intervals and resulting quantized-signal MAE annotations.

and recovers the imposed initial mismatch fastest. This correction capability does not make it universally robust: high current noise and burst dropout produce the largest HECM penalties because both state propagation and observer correction are affected when the causal input stream is corrupted or frozen. Its lookup-table dependence and flash footprint are additional deployment costs that must be balanced against its rapid initialization correction.

DD has the lowest nominal cell-macro error and the best adverse current-gain score in this comparison. It is also comparatively resilient to isolated sample loss and recovers fastest after burst dropout. Conversely, event-aligned voltage spikes expose a recurrent transient that full-window MAE hides. The result is specific to the frozen feature set and checkpoint; it indicates a channel-level vulnerability of this implementation, not a universal limitation of recurrent SOC estimators.

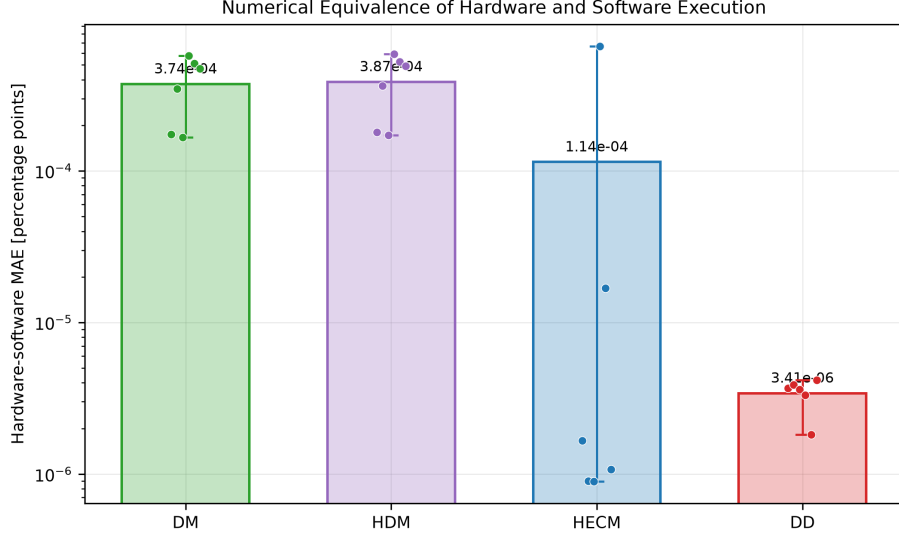


Figure 17: Numerical equivalence of STM32 and software execution over six cell sequences. Bars show mean hardware–software MAE in percentage points, dots show cells, and whiskers show the cell range. The logarithmic scale emphasizes that all implementation differences are negligible relative to model error.

5.2. Accuracy–performance trade-offs

No single scalar captures the observed trade-offs. DD leads the baseline MAE/RMSE ranking and the three illustrative weighted profiles, but HECM is substantially faster at correcting initialization mismatch and DM/HDM are orders of magnitude cheaper in inference time. Recovery itself is scenario dependent: HECM leads after initialization mismatch, whereas DD leads after burst dropout. Likewise, the reference-SOH ablation improves HDM and HECM but not DD. Model choice should therefore start from the relevant failure mechanism, latency budget, and state-service architecture rather than from one aggregate rank.

5.3. Implications for BMS deployment

For the tested LFP cells, DD is the strongest general-purpose accuracy/robustness candidate if its recurrent implementation and voltage-spike response are ac-

On-Device Inference-Latency Distributions Across Six Cells

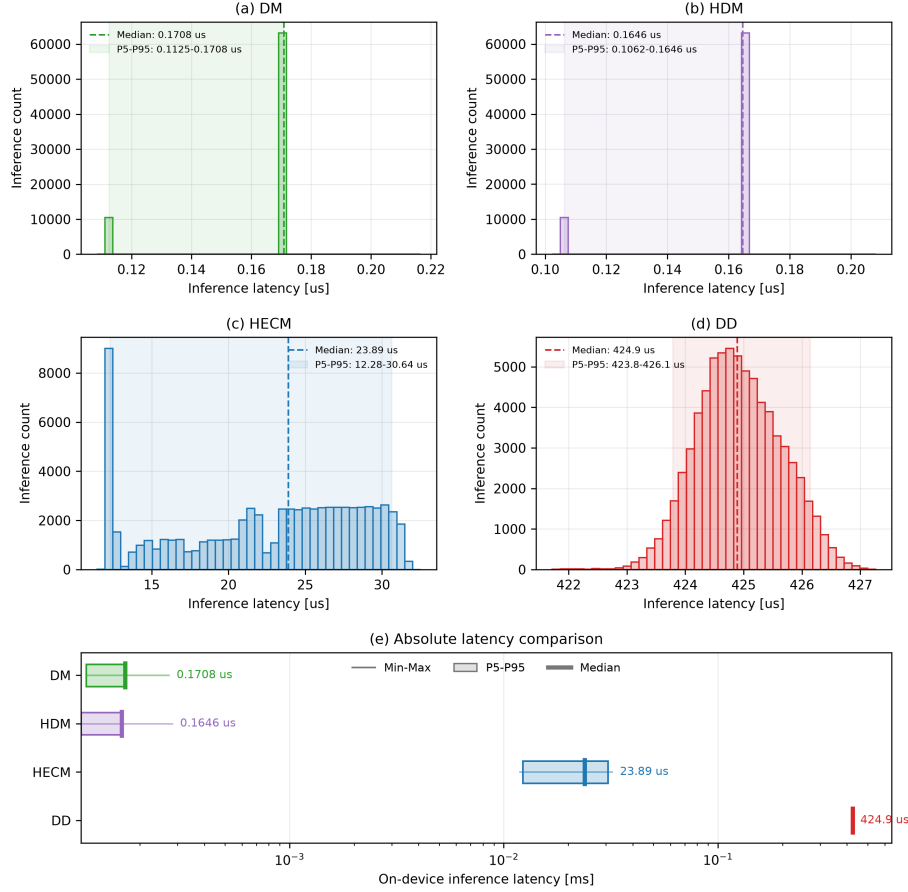


Figure 18: On-device SOC-core latency distributions across six cells and three replay rounds. Panels (a–d) show per-inference distributions; panel (e) compares minimum, 5th percentile, median, 95th percentile, and maximum on a logarithmic scale. DD denotes continuous-state inference in this cross-model latency comparison.

549 ceptable. HECM remains preferable when restart correction is critical and a
550 larger flash image is available. DM and HDM are attractive when deterministic
551 sub-microsecond execution and small memory dominate, but persistent current
552 calibration and communication continuity then require stronger system-level
553 safeguards. These recommendations apply to the tested models, not to their

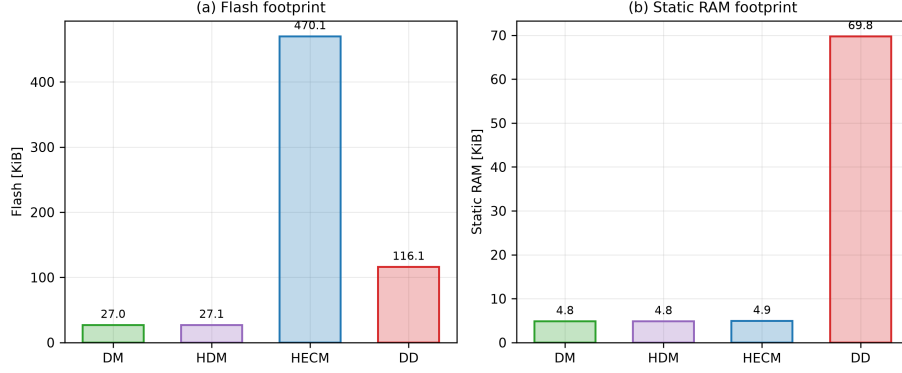


Figure 19: Measured release-image flash and static-RAM footprints for the isolated STM32 SOC implementations. Flash and RAM are firmware properties and are therefore reported once per implementation rather than as artificial cell-dependent distributions.

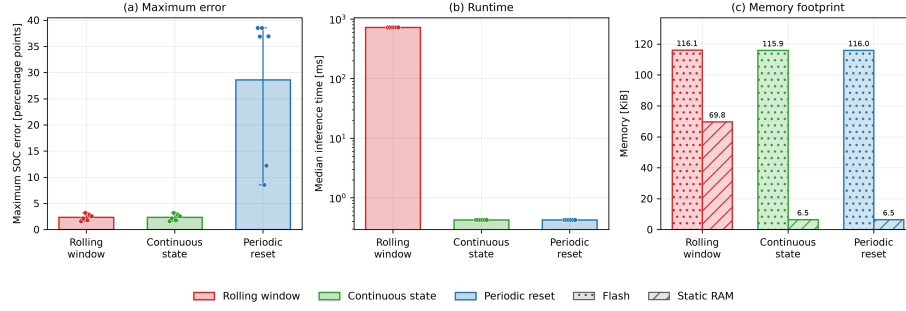


Figure 20: DD inference-mode trade-off on the STM32. The panels compare maximum dataset-SOC error, median latency, and firmware memory for exact rolling-window, continuous-state, and periodically reset recurrent execution. Cell points show that periodic resets, not continuous forwarding, create the dominant accuracy loss.

554 estimator families in the abstract.

555 The STM32 results narrow the gap between algorithmic and deployment
 556 claims. All four isolated SOC cores meet the one-second deadline with the
 557 tested implementation, but their margins differ by more than six orders of mag-
 558 nitude between DM/HDM and exact rolling-window DD. Continuous-state DD
 559 removes most of that cost, at the price of changing inference semantics. The
 560 hardware test does not include the shared SOH-LSTM, peak stack, energy, pack

Table 2: Decision-oriented recommendation map derived from the benchmark results. The table summarizes which estimator class is most suitable when one deployment priority dominates the design decision.

Primary deployment priority	Recommended class	Main supporting benchmark evidence	Main trade-off to consider
Lowest nominal SOC error	DD	Lowest six-cell macro MAE (0.0258) and RMSE (0.0300)	Rolling-window execution is the most computationally expensive implementation
Fastest recovery from initialization mismatch	HECM	Mean stable recovery of 1.03 h under the common criterion	Highest flash footprint and slowest post-dropout recovery
Lowest adverse current-gain penalty	DD	Δ MAE 0.0038 at 3%, below HECM, HDM, and DM	Local voltage-spike transient remains visible
Lowest periodic/random sample-loss penalty	DD	Δ MAE 0.0011/0.0026	One-hour burst dropout still increases MAE by 0.0134
Lowest deterministic compute cost	DM/HDM	Sub-microsecond median SOC-core latency and approximately 27 KiB flash	Integration structure is sensitive to gain error and missing charge information
Practical fast neural deployment	DD continuous state	Approximately 0.425 ms median and 6.5 KiB static RAM	Different recurrent semantics from the primary rolling-window benchmark

communication, or concurrent BMS tasks; therefore it supports SOC-core feasibility, not complete BMS schedulability. Table 2 summarizes the evidence without hiding these boundaries.

Figure 15 complements the recommendation map with relative scores, not additional measurements. Lower-is-better inputs are min–max normalized within the four tested implementations. Accuracy combines baseline MAE, RMSE, and 95th-percentile error; robustness combines cross-scenario penalties; and recovery combines initialization and burst-dropout behavior. The three profiles assign 60% to their named dimension and 20% to each remaining dimension. DD scores highest in all three illustrative profiles (0.933, 0.842, and 0.890), followed by HECM, HDM, and DM. This consistency reflects the present input metrics and normalization; it must not be generalized beyond the tested cells or interpreted as proof that DD dominates every individual scenario.

5.4. Limitations and future extensions

The study remains limited to six holdout cells from one controlled LFP dataset and therefore cannot establish transfer to NMC/NCA chemistry, other form factors, pack imbalance, other thermal environments, or field duty cycles.

578 The load groups are descriptive strata with $n = 2$ low-load, $n = 3$ middle-
 579 load, and one exploratory high-load cell; they are not randomized treatments.
 580 C27 contains no mid-life or aged window, and missing states are not imputed.
 581 With six independent cells, confidence intervals and effect magnitudes are more
 582 informative than dichotomized significance decisions.

583 Only measurement, timing, availability, and initialization channels are ma-
 584 nipulated. Cell/pack parameter mismatch, ECM aging drift, hysteresis er-
 585 ror, thermal gradients, sensor cross-sensitivity, balancing and contactor faults,
 586 communication-protocol failures, numerical faults, and deadline interference re-
 587 main outside the campaign. DD initialization is implemented through an SOC-
 588 equivalent Q_c offset rather than direct hidden-state corruption. The voltage-
 589 spike mechanism is supported by event alignment but not proven by an input-
 590 channel ablation. HECM results also depend on the quality and independence
 591 of its parameter lookup tables.

592 The hardware experiment isolates SOC cores and passes SOH as an external
 593 causal trace. It does not measure end-to-end SOH latency, peak stack, energy,
 594 CPU utilization under concurrent BMS tasks, or pack-level closed-loop behav-
 595 ior. Future work should therefore prioritize peak-memory and energy instru-
 596 mentation, scheduling tests with the SOH service, clustered-fault experiments,
 597 hidden-state ablations, and independent validation on additional chemistries
 598 and duty cycles.

599 6. Conclusion

600 This work replaces a clean-data, single-trajectory comparison with a causal
 601 six-cell LFP benchmark covering 19 measurement-only cases, repeated stochas-
 602 tic seeds, cell-aware uncertainty, and isolated STM32 validation. DD has the
 603 lowest nominal macro error and the lowest adverse current-gain penalty in the
 604 tested set. HECM corrects initialization mismatch fastest, DD recovers fastest
 605 after burst dropout, sample loss matters more than timing jitter, and event
 606 alignment exposes a DD voltage-spike transient that global MAE conceals. The

paired SOH ablation further shows that reference SOH does not uniformly improve every SOC pipeline.

The practical conclusion is scenario specific. DD provides the strongest aggregate result for these cells but requires careful recurrent deployment semantics and spike-aware validation. HECM offers rapid observer-based re-anchoring at higher flash cost, whereas DM and HDM provide exceptional timing and memory margins with greater dependence on current accuracy and data continuity. The STM32 results confirm close numerical equivalence and ample nominal timing margin for all isolated SOC cores, while leaving complete SOC-SOH scheduling, peak memory, and energy as open system-level validation tasks.

Appendix A. Holdout Coverage and Frozen Protocol

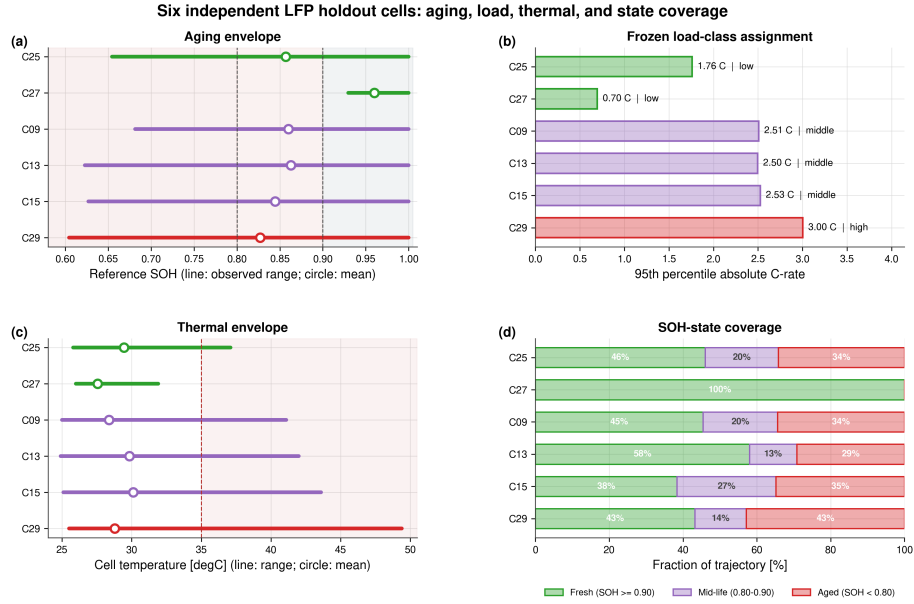


Figure A.21: Quantitative coverage of the six independent holdout cells. Panels show observed SOH, the descriptive load-class assignment, thermal envelope, and full-trajectory SOH-state coverage. High-load evidence consists only of C29 and is exploratory.

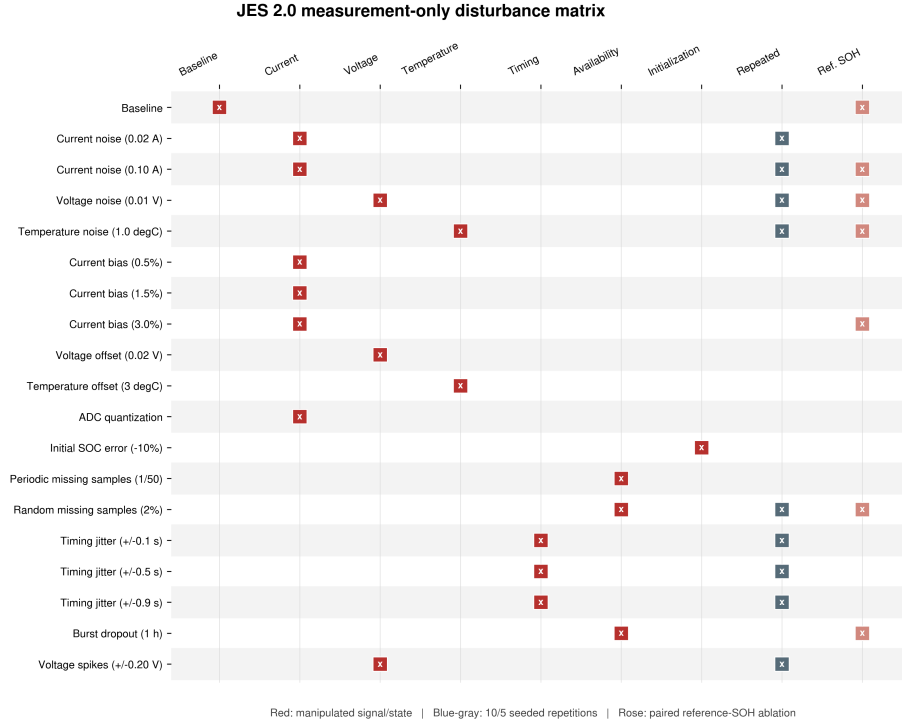


Figure A.22: Complete 19-case JES2 matrix. Red markers identify the manipulated measurement, timing, availability, or initialization channel; blue-gray markers denote repeated stochastic cases; rose markers identify paired reference-SOH ablations.

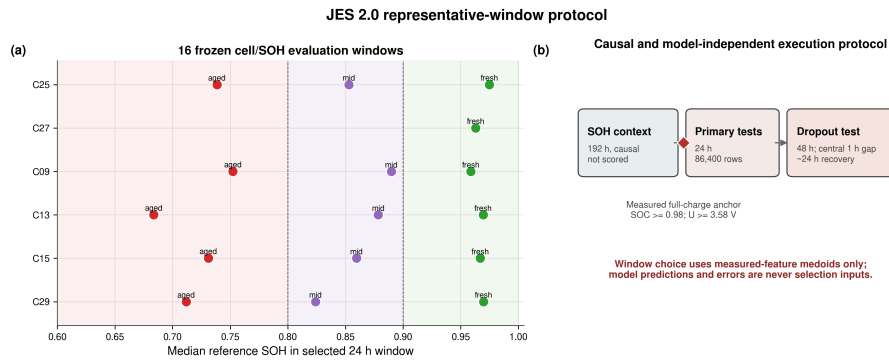


Figure A.23: Frozen representative-window protocol. Sixteen cell/SOH windows are selected using measured-feature medoids only. Each scored 24-h window follows a 192-h causal SOH context; the dropout case extends evaluation to 48 h.

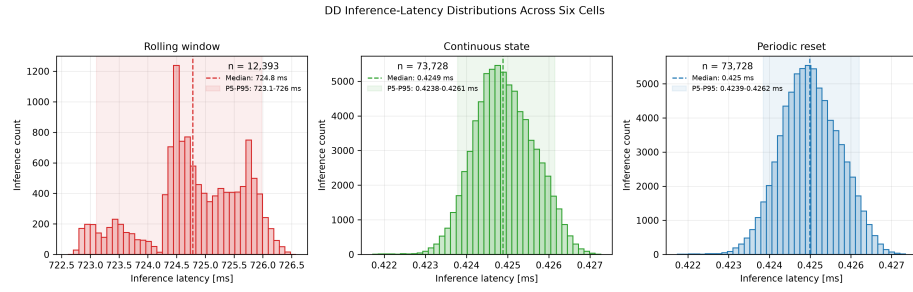


Figure A.24: DD latency distributions over the six hardware cell sequences. Exact rolling-window execution is shown separately from continuous-state and periodically reset execution because their scales and recurrent semantics differ substantially.

Table B.3: Six-cell baseline cell-macro statistics with hierarchical 95% confidence intervals.

Estimator	MAE [SOC]	RMSE [SOC]	P95 error [SOC]
DM	0.0678 [0.0491, 0.0809]	0.0733 [0.0524, 0.0873]	0.1086 [0.0754, 0.1307]
HDM	0.0405 [0.0297, 0.0506]	0.0438 [0.0317, 0.0555]	0.0648 [0.0439, 0.0845]
HECM	0.0335 [0.0246, 0.0429]	0.0389 [0.0289, 0.0496]	0.0646 [0.0467, 0.0812]
DD	0.0258 [0.0189, 0.0323]	0.0300 [0.0224, 0.0367]	0.0508 [0.0375, 0.0618]

Table B.4: Selected cell-macro scenario penalties relative to paired baseline. Current-gain values use the adverse sign from each matched $\pm 3\%$ pair.

Scenario	DM	HDM	HECM	DD
Current noise, $\sigma_I = 0.10$ A	-0.0009	+0.0000	+0.0056	-0.0000
Current gain, adverse 3%	+0.0107	+0.0100	+0.0059	+0.0038
ADC quantization	+0.0037	+0.0034	-0.0018	+0.0002
Random missing samples, 2%	+0.0079	+0.0080	+0.0038	+0.0026
Timing jitter, ± 0.9 s	-0.0003	+0.0003	+0.0006	+0.0001
Burst dropout, 1 h	+0.0067	+0.0134	+0.0221	+0.0134
Voltage spikes, ± 0.20 V	+0.0000	-0.0003	+0.0009	-0.0001

Table B.5: Recovery and target-hardware summary. Recovery values are equal-weight cell means.

Metric	DM	HDM	HECM	DD
Initial stable recovery [h]	22.28	22.28	1.03	5.57
Burst recovery/censor time [h]	5.13	5.67	12.45	4.13
Median SOC-core latency [μ s]	0.171	0.165	23.9	424.9 ^a
Flash [KiB]	27.0	27.1	470.1	116.1
Static RAM [KiB]	4.8	4.8	4.9	69.8

^aContinuous-state DD; exact rolling-window median is approximately 724 ms.

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623 **CRedit authorship contribution statement**

624 The author contribution statement is provided in the non-blinded submission
625 materials.

626 **Declaration of competing interest**

627 The authors declare that they have no known competing financial interests or
628 personal relationships that could have appeared to influence the work reported
629 in this paper.

630 **Declaration of generative AI and AI-assisted technologies in the manuscript**
631 **preparation process**

632 During the preparation of this work the author(s) used ChatGPT, DeepL,
633 and Grammarly in order to improve the linguistic quality and style of the
634 manuscript. After using these tools, the author(s) reviewed and edited the con-
635 tent as needed and take(s) full responsibility for the content of the published
636 article.

637 **Data availability**

638 The underlying laboratory dataset is the MGFarm LFP 18650 cyclic-aging
639 dataset described in [28]. The benchmark scripts and model runners are lo-
640 cated in `DL\_Models/LFP\_SOC\_SOH\_Model/4\_simulation\_environment`,
641 and the paper-facing figures and tables are collected under `DL\_Models/LFP\`

642 _SOC_SOH_Model/4_simulation_environment/results. The data and code
643 underlying this study will be made available on reasonable request.

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